CS 101 Lab 2 Revision Thu

For part 1, Update the question to input the ***Volume*** and the ***Base Edge length* (a)** of the pyramid.Using these values, you need to "reverse engineer" the pyramid dimensions:

1. Calculate and output the **vertical height (h)** of the pyramid.
    - *Hint: Rearrange the volume formula (V = a^2 * h / 3) to solve for the height (h). The derived formula is: h = (3 * V) / a^2*
2. Calculate and output the **slant height (s)**.
    - *Hint: Use Pythagoras with the height you just found: s = sqrt(h^2 + (a/2)^2).*
3. Calculate and output the **total surface area**.

Furthermore, modify your answer to Question 2 so that it also outputs the **difference** between the weight on Earth and the weight on each different planet in parentheses next to each weight.

- The difference should be calculated as: `(Weight on Earth - Weight on Planet)`
- Note that for planets with higher gravity than Earth (like Jupiter), the difference will be negative.
- The output format must strictly follow the alignment shown below.

Sample Run (Question 1):

```
Enter the volume of the pyramid: 400.0
Enter the base edge length: 10.0
The height of the pyramid is:                    12.0
The slant height of the pyramid is:              13.0
The total surface area is:                      360.0
```

Sample Run (Question 2):

```
Enter the mass of the first probe (kg): 100
Enter the mass of the second probe (kg): 55
                                   EARTH                   MOON                    MARS                JUPITER
     PROBE ONE(100.0kg)            980.7      162.0(    818.7)      371.1(    609.6)      2479.0( -1498.3)
     PROBE TWO(55.0kg)             539.4       89.1(    450.3)      204.1(    335.3)      1363.5(   -824.1)
```